Sahaj Bhadja

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Summary

Systems-minded software engineer pursuing an MSCS at USC with 2+ years building tested backend services, data pipelines, and performance-sensitive workflows. Strong CS fundamentals, C++/Python/Linux experience, networking basics, telemetry-style debugging, and interest in autonomous vehicle hardware integration.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Designed and optimized performance-sensitive backend workflows supporting 500+ users, reducing latency from **1.2s to 350ms** through profiling, query tuning, and service fixes.
- Built validation pipelines processing 5,000+ monthly jobs with reconciliation checks, CI/CD releases, and runbooks, improving telemetry-style reporting and production debugging.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed React and Flask services handling 10,000+ daily requests, implementing REST APIs, debugging backend behavior, and writing tests that improved release confidence.
- Built AI-assisted document workflows with OCR, LLM prompts, and AWS-backed persistence, reducing feedback turnaround by **60%** while documenting edge cases and integration failures.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built Python evaluation pipelines across 100,000+ records, producing reproducible traces, reports, and debugging artifacts that clarified model behavior and improved experimental reliability.
- Analyzed failure patterns across generated outputs and datasets, translating technical findings into actionable recommendations while strengthening validation coverage and engineering handoffs.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered API and ETL microservices using Python, AWS Glue, and S3, improving response time by **30%** while supporting scalable integrations and reliable production workflows.
- Maintained 20+ automated test suites using Jenkins, PyTest, and Mockito, increasing CI reliability by **68%**, reducing regressions, and accelerating defect isolation across services.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built a MERN-based platform with WebSockets and REST APIs, supporting 2M+ synchronized records and six workflows through modular services and responsive interfaces.
- Automated Docker-based build and test flows with Jenkins, cutting release time by **50%**, improving deployment consistency, and reducing friction during validation and troubleshooting.

Skills

Systems: C++, Python, Linux CLI, data structures, algorithms, memory-management fundamentals, profiling, scripting
Hardware & Networking: TCP/IP, UDP, serial concepts, sensor-data workflows, telemetry, CAN/ROS2 learning, embedded systems interest
Backend & Tools: REST APIs, Flask, FastAPI, PostgreSQL, MySQL, MongoDB, Git/GitHub, Docker, CI/CD, Jenkins
Validation: Unit testing, integration testing, HIL-style test interest, fault analysis, runbooks, safety-critical software mindset

Projects

CPU Scheduling Simulator Live | *C++, React.js, JavaScript*
- Implemented scheduling algorithms and visual comparisons for operating-system workloads, reinforcing C++ fundamentals, data structures, performance reasoning, and clear explanation of system behavior.

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Developed an NLP service pipeline with containerized deployment and CI/CD automation, improving reproducibility, API reliability, and troubleshooting speed for model-backed workflows.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*
- Built an AI-powered SQL workspace supporting 10+ databases, combining schema inspection, query generation, Dockerized services, and debugging flows for faster data validation.